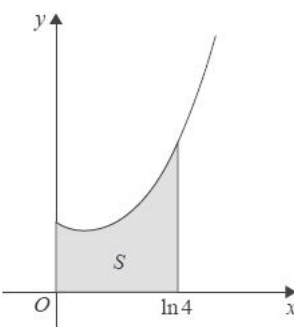


Given: Due:	Further Maths (Core Pure) HW 28	MARK SCHEME
Chapter 5: Volumes of Revolution		
Q1. The finite region S, shown shaded in Figure 2, is bounded by the y-axis, the x-axis, the line with equation $x = \ln 4$ and the curve with equation $y = e^x + 2e^{-x}, \quad x \geq 0$ The region S is rotated through 2π radians about the x-axis. Use integration to find the exact value of the volume of the solid generated. Give your answer in its simplest form. [Solutions based entirely on graphical or numerical methods are not acceptable.]		 Diagram not drawn to scale
		(7)

This question offered good discrimination across students of all abilities. Most students applied the volume

formula $\pi \int y^2 dx$ correctly to give a correct $\pi \int_0^{\ln 4} (e^x + 2e^{-x})^2 dx$, although a few used incorrect formulae such as $2\pi \int y^2 dx$, $\int y^2 dx$ or even $\int y dx$. Only a few students integrated $(e^x + 2e^{-x})^2$ wrongly at this stage to give terms such as $\frac{(e^x + 2e^{-x})^3}{3}$ or $\frac{(e^x + 2e^{-x})^3}{3(e^x - 2e^{-x})}$. The majority expanded $(e^x + 2e^{-x})^2$ correctly to give $e^{2x} + 4e^{-2x} + 4$, although some obtained incorrect expressions such as $e^{2x} + 4e^{-2x}$, $e^{2x} + 2e^{-2x} + 4$ or $e^{2x} + 4e^{-2x} + 4e$. Disappointingly, at this level, a few students expanded $(e^x + 2e^{-x})^2$ to give $e^{x^2} + 4e^{-x^2} + 4$. Most students integrated their correctly with only a small number of students differentiating this expression. Most students applied the limits of $\ln 4$ and 0 to their integrated expression and found an exact answer. Some students at this stage made sign errors, manipulation errors and in some cases omitted π from their final exact answer.

Question Number	Scheme	Notes	Marks
Way 1	$y = e^x + 2e^{-x}, x \geq 0$		
	$\{V = \pi \int_0^{\ln 4} (e^x + 2e^{-x})^2 dx$	For $\pi \int (e^x + 2e^{-x})^2$ Ignore limits and dx . Can be implied.	B1
	$= \{\pi\} \int_0^{\ln 4} (e^{2x} + 4e^{-2x} + 4) dx$	Expands $(e^x + 2e^{-x})^2 \rightarrow \pm \alpha e^{2x} \pm \beta e^{-2x} \pm \delta$ where $\alpha, \beta, \delta \neq 0$. Ignore π , integral sign, limits and dx . This can be implied by later work.	M1
	$= \{\pi\} \left[\frac{1}{2} e^{2x} - 2e^{-2x} + 4x \right]_0^{\ln 4}$	Integrates at least one of either $\pm \alpha e^{2x}$ to give $\pm \frac{\alpha}{2} e^{2x}$ or $\pm \beta e^{-2x}$ to give $\pm \frac{\beta}{2} e^{-2x}$, $\alpha, \beta \neq 0$	M1
		dependent on the 2 nd M mark $e^{2x} + 4e^{-2x} \rightarrow \frac{1}{2} e^{2x} - 2e^{-2x}$, which can be simplified or un-simplified	A1
		$4 \rightarrow 4x$ or $4e^0 x$	B1 cao
	$= \{\pi\} \left(\left(\frac{1}{2} e^{2(\ln 4)} - 2e^{-2(\ln 4)} + 4(\ln 4) \right) - \left(\frac{1}{2} e^{2(0)} - 2e^{-2(0)} + 4(0) \right) \right)$	dependent on the previous method mark. Some evidence of applying limits of $\ln 4$ o.e. and 0 to a changed function in x and subtracts the correct way round. Note: A proper consideration of the limit of 0 is required.	dM1
	$= \{\pi\} \left(\left(8 - \frac{1}{8} + 4 \ln 4 \right) - \left(\frac{1}{2} - 2 \right) \right)$		
	$= \frac{75}{8} \pi + 4\pi \ln 4$ or $\frac{75}{8} \pi + 8\pi \ln 2$ or $\pi \left(\frac{75}{8} + 4 \ln 4 \right)$ or $\pi \left(\frac{75}{8} + 8 \ln 2 \right)$ or $\frac{75}{8} \pi + \ln 2^{8\pi}$ or $\frac{75}{8} \pi + \pi \ln 256$ or $\ln \left(2^{8\pi} e^{\frac{75}{8}\pi} \right)$ or $\frac{1}{8} \pi (75 + 32 \ln 4)$, etc		A1 isw
			[7]

Question Notes			
Note	π is only required for the 1 st B1 mark and the final A1 mark.		
Note	Give 1 st B0 for writing $\pi \int y^2 dx$ followed by $2\pi \int (e^x + 2e^{-x})^2 dx$		
Note	Give 1 st M1 for $(e^x + 2e^{-x})^2 \rightarrow e^{2x} + 4e^{-2x} + 2e^0 + 2e^0$ because $\delta = 2e^0 + 2e^0$		
Note	A decimal answer of 46.8731... or $\pi(14.9201...)$ (without a correct exact answer) is A0		
Note	$\pi \left[\frac{1}{2} e^{2x} - 2e^{-2x} + 4x \right]_0^{\ln 4}$ followed by awrt 46.9 (without a correct exact answer) is final dM1A0		
Note	Allow exact equivalents which should be in the form $a\pi + b\pi \ln c$ or $\pi(a + b \ln c)$, where $a = \frac{75}{8}$ or $9\frac{3}{8}$ or 9.375. Do not allow $a = \frac{150}{16}$ or $9\frac{6}{16}$		
Note	Give B1M0M1A1B0M1A0 for the common response $\pi \int_0^{\ln 4} (e^x + 2e^{-x})^2 dx \rightarrow \pi \int_0^{\ln 4} (e^{2x} + 4e^{-2x}) dx = \pi \left[\frac{1}{2} e^{2x} - 2e^{-2x} \right]_0^{\ln 4} = \frac{75}{8} \pi$		
Question Number	Scheme	Notes	Marks
Way 2	$y = e^x + 2e^{-x}, x \geq 0$		
	$\{V\} = \pi \int_0^{\ln 4} (e^x + 2e^{-x})^2 dx$	For $\pi \int (e^x + 2e^{-x})^2$ Ignore limits and dx. Can be implied.	B1
	$u = e^x \Rightarrow \frac{du}{dx} = e^x = u$ and $x = \ln 4 \Rightarrow u = 4, x = 0 \Rightarrow u = e^0 = 1$		
	$V = \{\pi\} \int_1^4 \left(u + \frac{2}{u}\right)^2 \frac{1}{u} du = \{\pi\} \int_1^4 \left(u^2 + \frac{4}{u^2} + 4\right) \frac{1}{u} du$		
	$= \{\pi\} \int_1^4 \left(u + \frac{4}{u^3} + \frac{4}{u}\right) du$	$(e^x + 2e^{-x})^2 \rightarrow \pm \alpha u \pm \beta u^{-3} \pm \delta u^{-1}$ where $u = e^x, \alpha, \beta, \delta \neq 0$. Ignore π , integral sign, limits and du . This can be implied by later work.	M1
	$= \{\pi\} \left[\frac{1}{2} u^2 - \frac{2}{u^2} + 4 \ln u \right]_1^4$	Integrates at least one of either $\pm \alpha u$ to give $\pm \frac{\alpha}{2} u^2$ or $\pm \beta u^{-3}$ to give $\pm \frac{\beta}{2} u^{-2}, \beta \neq 0$, where $u = e^x$ dependent on the 2 nd M mark $u + 4u^{-3} \rightarrow \frac{1}{2} u^2 - 2u^{-2}$, simplified or un-simplified, where $u = e^x$	M1 A1
	$= \{\pi\} \left(\left(\frac{1}{2} (4)^2 - \frac{2}{(4)^2} + 4 \ln 4 \right) - \left(\frac{1}{2} (1)^2 - \frac{2}{(1)^2} + 4 \ln 1 \right) \right)$	dependent on the previous method mark. Some evidence of applying limits of 4 and 1 to a changed function in u [or $\ln 4$ o.e. and 0 to an integrated function in x] and subtracts the correct way round.	dM1
	$= \{\pi\} \left(\left(8 - \frac{1}{8} + 4 \ln 4 \right) - \left(\frac{1}{2} - 2 \right) \right)$		
	$= \frac{75}{8} \pi + 4\pi \ln 4$ or $\frac{75}{8} \pi + 8\pi \ln 2$ or $\pi \left(\frac{75}{8} + 4 \ln 4 \right)$ or $\pi \left(\frac{75}{8} + 8 \ln 2 \right)$ or $\frac{75}{8} \pi + \ln 2^{8\pi}$ or $\frac{75}{8} \pi + \pi \ln 256$ or $\ln \left(2^{8\pi} e^{\frac{75}{8}\pi} \right)$ or $\frac{1}{8} \pi (75 + 32 \ln 4)$, etc		A1 isw
			[7]

Q2.

$$f(x) = 2x^{\frac{1}{3}} + x^{-\frac{2}{3}} \quad x > 0$$

The finite region bounded by the curve $y = f(x)$, the line $x = \frac{1}{8}$, the x -axis and the line $x = 8$ is rotated through θ radians about the x -axis to form a solid of revolution.

Given that the volume of the solid formed is $\frac{461}{2}$ units cubed, use algebraic integration to find the angle θ through which the region is rotated.

(Total for question = 8 marks)

This question was generally answered well, though there was a significant minority who did not offer any attempt at all. Most candidates realised that they would need to find the volume of revolution and then scale it to find the angle but had difficulty identifying the correct scaling.

A few did not realise they needed to find y^2 and consequently made little progress. Others did attempt the squaring, but with poor algebra, resulting in only two terms, or incorrect powers, though these were still able to gain the marks for the overall strategy and some for attempting the integration. Most, however, did manage to accurately expand and integrate.

Integration of fractional indices was generally very well done, even if the original expansion was not correct, and most would attempt the correct substitution of limits before attempting scaling. Use of a calculator to evaluate the integral was fortunately rare, as it was costly given that the question specified algebraic integration must be used.

The final two marks were useful discriminator marks, requiring a fully correct strategy to find the required angle. Roughly one fifth of candidates successfully achieved a correct answer, though a correct strategy following errors was seen numerous times too. However many candidates used $\theta \int y^2 dx$ instead of $\frac{\theta}{2} \int y^2 dx$ for the volume and this led to a common incorrect answer of $\frac{22}{9}$. Other incorrect scaling attempts were made, and it was more common to first find the volume of rotation through 2π before attempting to scale this volume in some way.

Question	Scheme	Marks	AOs
	A correct overall strategy, an attempt at integrating y^2 with respect to x combine in some way with the volume of revolution formula (use of $\pi \int y^2 dx$ or $\alpha \int y^2 dx$ for any variable α is fine) followed by attempt to find an angle/form an equation in θ	M1	3.1a
	$y^2 = kx^{\frac{2}{3}} + \dots + \frac{m}{x^{\frac{2}{3}}}$ or $y^2 = kx^{\frac{2}{3}} + \dots + mx^{-\frac{2}{3}}$ where $\dots$ is one or two more terms.	M1	1.1b
	$y^2 = 4x^{\frac{2}{3}} + 4x^{-\frac{2}{3}} + x^{-\frac{4}{3}}$ or $y^2 = 4x^{\frac{2}{3}} + 2x^{-\frac{2}{3}} + x^{-\frac{4}{3}} + 2x^{-\frac{2}{3}}$ (oe)	A1	1.1b
	$\int y^2 dx = \int 4x^{\frac{2}{3}} + \frac{4}{x^{\frac{2}{3}}} + \frac{1}{x^{\frac{4}{3}}} dx = \alpha x^{\frac{5}{3}} + \beta x^{\frac{1}{3}} + \gamma x^{-\frac{1}{3}}$	M1	1.1b
	$= \frac{12x^{\frac{5}{3}}}{5} + 6x^{\frac{1}{3}} - \frac{3}{x^{\frac{1}{3}}}$ (oe)	A1ft A1	1.1b 1.1b
	$\frac{\theta}{2} \left[\frac{12x^{\frac{5}{3}}}{5} + 6x^{\frac{1}{3}} - \frac{3}{x^{\frac{1}{3}}} \right]_{\frac{1}{8}}^8 = \frac{461}{2}$ $\Rightarrow \frac{\theta}{2} \left(\frac{12 \times 8^{\frac{5}{3}}}{5} + 6 \times 8^{\frac{1}{3}} - \frac{3}{8^{\frac{1}{3}}} \right) - \left(\frac{12 \times (\frac{1}{8})^{\frac{5}{3}}}{5} + 6 \times (\frac{1}{8})^{\frac{1}{3}} - \frac{3}{(\frac{1}{8})^{\frac{1}{3}}} \right) = \frac{461}{2} \Rightarrow \theta = \dots$ $\text{OR } \pi \left[\frac{12x^{\frac{5}{3}}}{5} + 6x^{\frac{1}{3}} - \frac{3}{x^{\frac{1}{3}}} \right]_{\frac{1}{8}}^8 = \pi \left(\frac{12 \times 8^{\frac{5}{3}}}{5} + 6 \times 8^{\frac{1}{3}} - \frac{3}{8^{\frac{1}{3}}} \right) - \left(\frac{12 \times (\frac{1}{8})^{\frac{5}{3}}}{5} + 6 \times (\frac{1}{8})^{\frac{1}{3}} - \frac{3}{(\frac{1}{8})^{\frac{1}{3}}} \right) = \dots$ followed by $\frac{\theta}{2\pi} \times \dots = \frac{461}{2} \Rightarrow \theta = \dots$	M1	3.1a
	$\theta = \frac{40}{9}$ (radians)	A1	1.1b
		(8)	
	(8 marks)		



Notes

M1	A correct overall strategy, either finding full volume rotated by 2π first, then performing some kind of scaling, or using $\alpha \int y^2 dx$ for a variable α (ideally $\frac{\theta}{2}$, but for the strategy accept with any variable multiple), to form an equation in just the angle.
M1	Attempting to square y to a three or four term expression. Look for correct powers on first and last term with some term(s) in the middle.
A1	Correct expansion in three or four terms – award when first seen.
M1	Integrates y^2 w.r.t. x . Must have at least two terms in their y^2 with fractional indices. Power to be increased by 1 in at least two terms.
A1ft	Two terms of integral correct. Follow through on their expansion. Need not be simplified.
A1	Fully correct integral. Need not be simplified. May still be four terms
M1	Either : Substitutes limits and subtracts correct way round (must be seen or implied by the answer), and equates to $\frac{461}{2}$ if using $\frac{1}{2}\theta \int y^2 dx$ and proceeds to find θ . Or : Substitutes limits and subtracts correct way round (seen or implied) and multiplies by π to get the full volume AND then multiplies the result by $\frac{\theta}{2\pi}$ before equating to $\frac{461}{2}$.
A1	The method must be correct for this mark – so they must be using $\frac{\theta}{2} \int y^2 dx$ directly or $\pi \int y^2 dx$ and scale by $\frac{\theta}{2\pi}$ when setting equal to $\frac{461}{2}$
A1	Correct angle found. Accept $\frac{40}{9}$, awrt 4.44 or awrt 255° (as long as the degrees units are made clear – do not accept just 255) isw once a correct value of θ is found.

Special case The question specified that algebraic integration must be used, so use of a calculator to find the integral cannot score the marks for integration but may be allowed the strategy and answer marks. A maximum of M1M0A0M0A0A0M1A1 is available in such cases.

Expanding y^2 first but showing no integration can score the second M and first A (if earned) as well.

Note that $\int_{1/8}^8 (2x^{1/2} + x^{-1/2})^2 dx = \frac{4149}{40} = 103.725$ but just this alone is worth **no marks**. There must

be an attempt to incorporate this within a strategy to gain access to marks.

Q3.

Diagrams not drawn to scale.

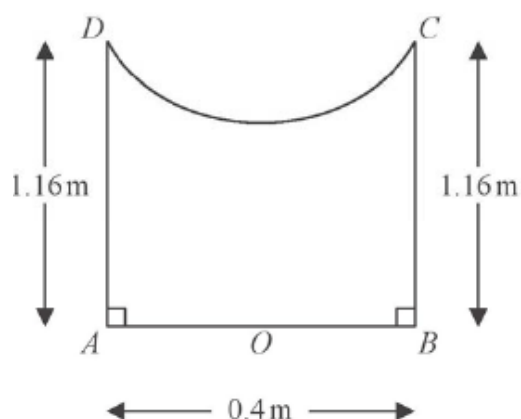


Figure 1

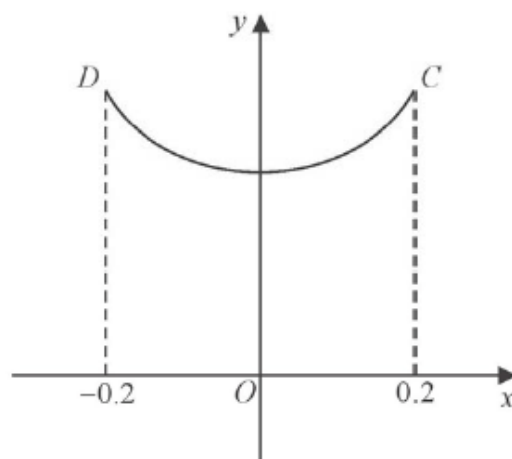


Figure 2

Figure 1 shows the central cross-section $AOB CD$ of a circular birdbath, which is made of concrete. Measurements of the height and diameter of the birdbath, and the depth of the bowl of the birdbath have been taken in order to estimate the amount of concrete that was required to make this birdbath.

Using these measurements, the cross-sectional curve CD , shown in Figure 2, is modelled as a curve with equation

$$y = 1 + kx^2, \quad -0.2 \leq x \leq 0.2,$$

where k is a constant and where O is the fixed origin.

The height of the bird bath measured 1.16 m and the diameter, AB , of the base of the birdbath measured 0.40 m, as shown in Figure 1.

- (a) Suggest the maximum depth of the birdbath. (1)
- (b) Find the value of k . (2)
- (c) Hence find the volume of concrete that was required to make the birdbath according to this model. Give your answer, in m^3 , correct to 3 significant figures. (7)
- (d) State a limitation of the model. (1)

It was later discovered that the volume of concrete used to make the birdbath was 0.127 m^3 correct to 3 significant figures.

- (e) Using this information and the answer to part (c), evaluate the model, explaining your reasoning. (1)



Question	Scheme of Work PRACTICE		Marks	AOs
7(a)	Depth = 0.16 (m)		B1	2.2b
			(1)	
(b)	$y = 1 + kx^2 \Rightarrow 1.16 = 1 + k(0.2)^2 \Rightarrow k = \dots$		M1	3.3
	$\Rightarrow k = 4$ cao {So $y = 1 + 4x^2$ }		A1	1.1b
			(2)	
(c)	$\frac{\pi}{4} \int (y-1) dy$	$\frac{\pi}{4} \int y dy$	B1ft	1.1a
	$= \left\{ \frac{\pi}{4} \right\} \int_1^{1.16} (y-1) dy$	$= \left\{ \frac{\pi}{4} \right\} \int_0^{0.16} y dy$	M1	3.3
	$= \left\{ \frac{\pi}{4} \right\} \left[\frac{y^2}{2} - y \right]_1^{1.16}$	$= \left\{ \frac{\pi}{4} \right\} \left[\frac{y^2}{2} \right]_0^{0.16}$	M1	1.1b
	$= \frac{\pi}{4} \left(\left(\frac{1.16^2}{2} - 1.16 \right) - \left(\frac{1}{2} - 1 \right) \right) \{ = 0.0032\pi \}$	$= \frac{\pi}{4} \left(\left(\frac{0.16^2}{2} \right) - (0) \right) \{ = 0.0032\pi \}$	A1	1.1b
	$V_{\text{cylinder}} = \pi(0.2)^2(1.16) \{ = 0.0464\pi \}$		B1	1.1b
	Volume = $0.0464\pi - 0.0032\pi \{ = 0.0432\pi \}$		M1	3.4
	$= 0.1357168026\dots = 0.136(\text{m}^3) (3\text{sf})$		A1	1.1b
			(7)	
(d)	Any one of e.g. the measurements may not be accurate the inside surface of the bowl may not be smooth there may be wastage of concrete when making the bird bath		B1	3.5b
			(1)	
(e)	Some comment consistent with their values. We do need a reason e.g. $\left[\left(\frac{0.136 - 0.127}{0.127} \right) \times 100 = 7.0866\dots \right]$ so not a good estimate because the volume of concrete needed to make the bird bath is approximately 7% lower than that predicted by the model or We might expect the actual amount of concrete to exceed that which the model predicts due to wastage, so the model does not look suitable since it predicts more concrete than was used		B1ft	3.5a
			(1)	

**Question 7 notes:**

EXAM PAPERS PRACTICE

(a)**B1:** Infers that the maximum depth of the bird bath could be 0.16 (m)**(b)****M1:** Substitutes $y = 1.16$ and $x = 0.2$ or $x = -0.2$ into $y = 1 + kx^2$ and rearranges to give $k = \dots$ **A1:** $k = 4$ cao**(c)****B1ft:** Uses the model to obtain either $\frac{\pi}{(\text{their } k)} \int (y-1) dy$ or $\frac{\pi}{(\text{their } k)} \int y dy$ **M1:** Chooses limits that are appropriate to their model**M1:** Integrates y (with respect to y) to give $\pm \lambda y^2$, where $\lambda \neq 0$ is a constant**A1:** Uses their model correctly to give either $y-1 \rightarrow \frac{y^2}{2} - y$ or $y \rightarrow \frac{y^2}{2}$ **B1:** $V_{\text{cylinder}} = \pi(0.2)^2(1.16)$ or 0.0464π or $\frac{29}{625}\pi$, o.e.**M1:** Depends on **both** previous M marksUses the model to find $V_{\text{their cylinder}}$ – their integrated volume**A1:** 0.136 cao**(d)****B1:** States an acceptable limitation of the model**(e)****B1ft:** Compares the actual volume with their answer to (c). Makes an assessment of the model. E.g. evaluates the percentage error and uses this to make a sensible comment about the model with a reason

Q4.

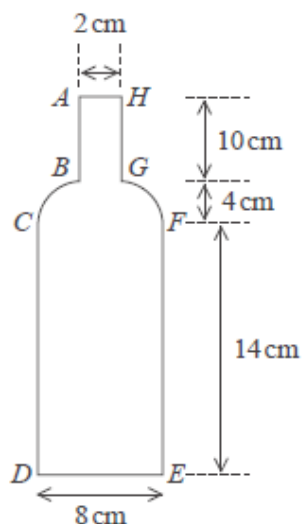


Figure 1

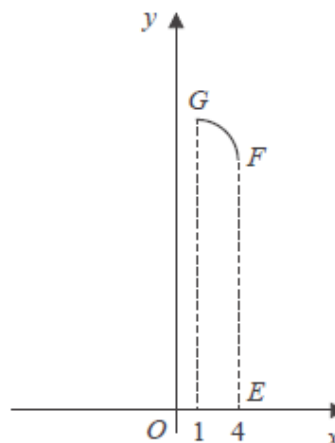


Figure 2

A mathematics student is modelling the profile of a glass bottle of water. Figure 1 shows a sketch of a central vertical cross-section $ABCDEFGHGA$ of the bottle with the measurements taken by the student.

The horizontal cross-section between CF and DE is a circle of diameter 8 cm and the horizontal cross-section between BG and AH is a circle of diameter 2 cm.

The student thinks that the curve GF could be modelled as a curve with equation

$$y = ax^2 + b, \quad 1 \leq x \leq 4,$$

where a and b are constants and O is the fixed origin, as shown in Figure 2.

(a) Find the value of a and the value of b according to the model. (2)

(b) Use the model to find the volume of water that the bottle can contain. (7)

(c) State a limitation of the model. (1)

The label on the bottle states that the bottle holds approximately 750 cm^3 of water.

(d) Use this information and your answer to part (b) to evaluate the model, explaining your reasoning. (1)

Question	Scheme	Marks	AOs
9(a)	$(4, 14), (1, 18) \Rightarrow 14 = a(4)^2 + b, 18 = a(1)^2 + b \Rightarrow a = \dots, b = \dots$	M1	3.3
	$a = -\frac{4}{15}, b = \frac{274}{15}$	A1	1.1b
		(2)	
(b)	$\pi \times 4^2 \times 14$ and $\pi \times 1^2 \times 10$	B1	1.1b
	$\pi \int x^2 dy = \frac{\pi}{4} \int (274 - 15y) dy$	B1ft	1.1a
	$= \frac{\pi}{4} \int_{14}^{18} (274 - 15y) dy$	M1	3.3
	$= \frac{\pi}{4} \left[274y - \frac{15y^2}{2} \right]_{14}^{18}$	M1 A1	1.1b 1.1b
	$V = 234\pi + \frac{\pi}{4} \left[274(18) - \frac{15(18)^2}{2} - \left(274(14) - \frac{15(14)^2}{2} \right) \right]$	ddM1	3.4
	$V = 268\pi \approx 842 \text{ cm}^3$	A1	2.2b
		(7)	



(c)	Any one of e.g. The measurements may not be accurate The equation of the curve may not be a suitable model The bottom of the bottle may not be flat The thickness of the glass may not have been considered The glass may not be smooth This part asks for a limitation of the model so their answer must refer to e.g. : • The measuring of the dimensions • The model used for the curve • The simplified model (the thickness of glass, the simplified shape, smoothness of the glass etc.)	B1	3.5b
		(1)	
(d)	There are 2 criteria for this mark: • A comparison of their value to 750 e.g. larger, smaller, about the same or a difference demonstrated e.g. $810 - 750 = \dots$ but not just a percentage error or just a difference with no calculation • A conclusion that is consistent with their values e.g. this is not a good model, this is a good model etc. If they reach an answer that is less than 750, they need to conclude that it is not a good model If they reach an answer that is greater than 750 then look for a sensible comment that is consistent with their value	B1ft	3.5a
		(1)	

(11 marks)

Notes
(a) M1: Chooses (4, 14) and (1, 18) and substitutes into the equation modelling the curve to obtain at least one correct equation and attempts to find the values of a and b. A1: Infers from the data in the model, the values of a and b (b) B1: Correct expressions for the 2 cylindrical parts. May be seen as a sum or as separate cylinders. B1ft: Uses the model to obtain $\pi \int \left(\frac{y - \text{their } b}{\text{their } a} \right) dy$ (Note that the π may be recovered later) M1: Chooses limits appropriate to the model i.e. 14 and 18 M1: Integrates to obtain an expression of the form $\alpha y + \beta y^2$ A1: Uses their model correctly to give $274y - \frac{15y^2}{2}$ ddM1: Uses the model to find the sum of their cylinders + their integrated volume. Must be a fully correct method here and is dependent on both previous method marks. So must have attempted the volumes of the cylinders "AHBG" and "CFED" and adds these to the magnitude of their integrated volume. A1: 268π or awrt 842 (c) B1: States an acceptable limitation of the model with no contradictory statements. (This is independent of part (b)) (d) B1ft: Compares the actual volume to their answer to part (b) and makes an assessment of the model with a reason with no contradictory statements.